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nf-core/funcscan: A Nextflow pipeline to identify the biosynthetic potential, resistome, and carbohydrate metabolism of bacterial (meta)genomes

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DOI: 10.xxxxxx/draft

Software

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Editor:

Submitted: 30 October 2026
Published: unpublished

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Summary

Genome-mining of bacterial DNA enables the discovery of antimicrobial resistance-related genes, genes required for the biosynthesis of low molecular weight natural products, and other specialised metabolites. However, execution of the multiple bioinformatic tools used in screening analyses remains inefficient due to heterogeneous software interfaces, reporting, and formatting of the output files of similar tools, which limits scalability of such analyses.

nf-core/funcscan is a portable and reproducible open source Nextflow bioinformatics pipeline for the screening of microbial functional features from assembled contigs or genomes. The pipeline executes up to 13 tools to simultaneously identify biosynthetic gene clusters, antimicrobial peptides, antibiotic resistance genes, and carbohydrate-active enzymes, while also performing taxonomic classification of partial or full genomes. To facilitate efficient results comparison and evaluation, it supports cross-tool output file standardisation and aggregation.

Statement of need

Researchers often use multiple tools to maximize detection sensitivity during genomic screening for potential gene candidates, as each tool uses different search algorithms and microbial

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metabolite databases. However, heterogeneous installations, inputs, and execution interfaces of these standalone tools impede scalability, and decrease reproducibility due to user-error when executed manually. Additionally, output formats are not standardised, making cross-comparison of results between tools and databases difficult and requiring inefficient manual postprocessing and inspection.

The current need for manual execution and postprocessing impacts the discovery of new drugs. The rise of multidrug resistant bacteria (Perry et al., 2016; Rascovan et al., 2016; Ventola, 2015) and reduction of global surveillance in healthcare and agriculture pose major threats to global health (Murray et al., 2022; World Health Organization, 2022). New antibiotics derived from low molecular weight bacterial natural products are needed to address these threats, but the discovery rate of novel molecules has declined as the rate of rediscovery has increased, necessitating the screening of ever larger metagenomic datasets. High-throughput and scalable approaches are needed to meet this challenge.

Here, we present nf-core/funcscan, a Nextflow (Di Tommaso et al., 2017) pipeline following nf-core best practices (Ewels et al., 2020; Langer et al., 2025) for the automation of in-parallel screening of biosynthesis, resistance, and carbohydrate metabolism genes using multiple tools and databases. The pipeline currently supports detection of biosynthetic gene clusters (BGCs), antimicrobial peptide (AMPs) genes, antimicrobial resistance genes (ARGs), and carbohydrate-active enzyme genes (CAZymes), as well as their gene clusters (CGCs).

State of the field

Previous efforts to scale up the predictive power of different tools for functional gene prediction include mettannotator (Gurbich et al., 2025), bacannot (Almeida et al., 2023), PathoFact (De Nies et al., 2021) SqueezeMeta (Tamames & Puente-Sánchez, 2019), MetaErg (Dong & Strous, 2019), and ARGs-OAP (Yin et al., 2022) (Table 1). However, these tools and pipelines serve different goals, such as focusing on individual gene categories or functions (e.g., antimicrobial resistance) or aiming to be 'end-to-end' pipelines that do not efficiently fit into existing workflows, for example by applying preferred read preparation and assembly steps. Furthermore, some pipelines do not provide sufficient contextual information (e.g., taxonomy) for key downstream laboratory analyses.

In particular, the main factors that distinguish nf-core/funcscan from the most similar pipeline, mettannotator, are: support for metagenomic assembly input rather than just genomes; automated taxonomic classification of contigs; more tools for ARG screening; support for AMP screening; and confirmed execution on other infrastructure than HPCs. Genomic functions which nf-core/funcscan does not screen for due to its focus on BGCs, AMPs, ARGs, and CAZymes/CGCs but mettannotator does are CRISPR arrays, antiphage defence, non-coding RNA, and pseudogenes.

Extensive command-line knowledge and manual installation of software dependencies are typically required to run many of the existing pipelines. This can make use of these tools difficult for biochemists and biologists who may have limited computational training. nf-core/funcscan aims to streamline this process by providing a single software in which to run all tools, perform screening from already assembled sequences, and provide aggregation of the screening results, with customizable options for execution (Table 1).

Workflow overview

nf-core/funcscan simultaneously predicts BGCs, AMPs, ARGs, and CAZymes/CGCs from partial or full (meta)genomic sequences. Output files from the tools of each of the categories are aggregated and standardised for easy cross-comparison (Fig. 1).

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Table 1: Comparison of nf-core/funcscan with other related pipelines for ARG, AMP, and BGC discovery. Parentheses indicate either unspecific gene screening or partly fulfilled criteria.

Feature	funcscan	mettannotator	bacannot	PathoFact	SqueezeMeta	MetaERG	ARGs-OAP
ARG screening	+	+	+	+	(+)	(+)	+
AMP screening	+	−	−	−	(+)	(+)	−
BGC screening	+	+	−	−	(−)	(−)	−
CAZyme screening	+	+	−	−	−	−	−
Taxonomic assignment of contigs	+	−	−	(−)	+	+	−
Output format standardisation	+	+	(+)	+	+	+	−
Container support (Docker, Singularity)	+	+	+	−	−	+	(−)
One-click installation	+	+	+	−	(−)	−	−
Local installation possible	+	+	+	+	+	+	−
Web-based execution possible	(+)	(+)	(+)	−	−	−	−
Community code review	+	+	−	−	−	−	−
Automated unit tests	+	+	(−)	(−)	−	−	−
License	MIT	Apache-2.0	GPL-3.0	GPL-3.0	GPL-3.0	AFL	AFL

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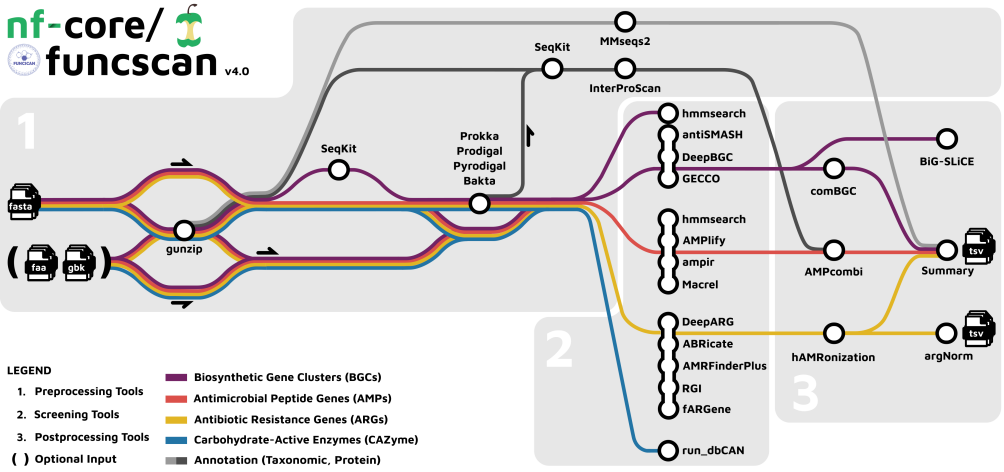


Figure 1: Workflow overview of nf-core/funcscan as of v4.0. (1) Genomic sequences are prepared and annotated with one of four open reading frame annotation tools. Two additional classification workflows can be used to classify contigs taxonomically (light gray) or obtain additional protein domain information (dark gray). (2) Depending on user-choice, the biosynthetic gene clusters (BGC, purple), antimicrobial peptide genes (AMP, orange), antibiotic resistance genes (ARG, yellow), or carbohydrate-active enzymes (CAZyme, blue) workflows with their customisable parameters are executed. (3) The results of all tools for each gene category are aggregated and saved in a human- and machine-readable tabular format.

Input preprocessing and open reading frame annotation

The pipeline takes a two- to five-column tabular sample-sheet as input (comma-separated, CSV format). This sample-sheet contains sample names, paths to (meta)genomic FASTA files, and optionally pre-generated amino-acid FASTA, GFF, or GBK format annotation files.

Preprocessing steps reduce runtime by removing too-short sequences with Seqkit (Shen et al., 2024), when they may produce no biologically meaningful results. Open reading frames are optionally predicted from the preprocessed sequences by one of four prokaryotic annotation tools: Bakta (Schwengers et al., 2021), Prodigal (Hyatt et al., 2010), Prokka (Seemann, 2014), and Pyrodigal (Larralde, 2022).

The pipeline downloads required screening-tool databases automatically for the user, and makes them available for future pipeline runs to minimise runtime and network traffic.

Gene screening and taxonomic classification

Users choose to screen genomic sequences in parallel with up to four dedicated subworkflows for BGCs, AMPs, ARGs, and CAZymes. Up to a total of 13 gene identification tools can be applied:

- **BGCs**: antiSMASH (Blin et al., 2025), DeepBGC (Hannigan et al., 2019), GECCO (Carroll et al., 2021), hmmsearch (Eddy, 2011)
- **ARGs**: ABRicate (Seemann, 2020), AMRFinderPlus (Feldgarden et al., 2019, 2021), DeepARG (Arango-Argoty et al., 2018), fARGene (Berglund et al., 2019), RGI (Alcock et al., 2023)
- **AMPs**: ampir (Fingerhut et al., 2021), AMPlify (Li et al., 2022, 2023), hmmsearch, Macrel (Santos-Júnior et al., 2020)
- **CAZymes**: run_dbCAN (Zheng et al., 2023)

To provide users information about potentially suitable bacterial hosts for downstream experiments (e.g., heterologous expression systems (Porse et al., 2018)), an additional optional parallel workflow can be activated to taxonomically classify input contigs with MMSeqs2

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(Mirdita et al., 2021). Optionally, generic protein domains and families can be further annotated with InterProScan (Jones et al., 2014).

Pipeline parameters can be adjusted by user-written or nf-core GUI-generated (https://nf-co.re/launch) Nextflow parameter files, or command-line arguments.

Aggregation of screening results

nf-core/funcscan integrates dedicated tools to aggregate and standardise heterogeneous output formats of multiple screening tools into a single human- and machine-readable table in CSV format per gene type. nf-core/funcscan uses a custom script 'comBGC' for BGC tool output aggregation, hAMRonization (Mendes et al., 2024) for ARGs, and AMPcombi (Herbst et al., 2025) for AMPs. These summaries are finally optionally complemented with results from the taxonomic classification workflow.

Building on the aggragation of screening results, two optional post-processing steps can be executed for the BGC and ARG workflows. BGCs predicted by antiSMASH and GECCO can be clustered into Gene Cluster Families (GCFs) by BiG-SLiCE (Kautsar et al., 2021) to enable comparative analysis of biosynthetic diversity across samples. The ARG summary provided by hAMRonization can be further normalised and mapped to the antibiotic resistance ontology (ARO) by argNorm (Ugarcina Perovic et al., 2025). This enhances ARG annotation by categorising drugs that ARGs confer resistance to.

Reproducibility and scalability

All nf-core pipelines utilise software environments (from the Bioconda project, Grüning et al., 2018) or containers (e.g., Docker, Singularity, primarily from the Biocontainers project, Veiga Leprevost et al., 2017) for each integrated tool. This provides the advantage of isolating the dependencies of all workflows from each other, thereby reducing installation problems. The pipeline is thus easy to install with minimal dependencies - Nextflow itself, and one of Nextflow-supported container/software environment management systems. For further portability, nf-core provides integrated configurations for more than 150 institutional computational infrastructures (e.g., HPCs) via nf-core/configs (https://nf-co.re/configs). Users on these infrastructures thus can run the pipelines with no further setup via a single parameter.

Research impact statement

nf-core/funcscan has an active user community of scientific users and developers on the public nf-core Slack and GitHub (https://nf-co.re/join). For example, the pipeline received the contribution of the CAZyme screening from community members outside of the original developers. User discussions and support on the pipeline and on related research topics occur on the nf-core Slack workspace. This illustrates the public interest and proactive efforts from scientific users to use, maintain, and improve the pipeline. The pipeline is also already actively being used in research, e.g., employed by Tighe et al. (2024), Janák et al. (2026), İstanbullugil et al. (2026), and Liepa et al. (2026).

AI usage disclosure

No generative AI tools were used in the development of this software, the writing of this manuscript, or the preparation of supporting materials.

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Acknowledgements

We thank Vedanth Ramji for adding argNorm to the ARG subworkflow. A full list of nf-core community members is available at https://nf-co.re/contributors. We thank Martin Klapper and Rosa Herbst for helpful feedback on relevant BGC and AMP properties during comBGC and AMPcombi development. J.F. received a fellowship from the International Leibniz Research School (under the head of the Jena School for Microbial Communication, JSMC). This work was conducted while J.F. was affiliated with the Department of Paleobiotechnology, Leibniz Institute for Natural Product Research and Infection Biology Hans Knöll Institute, Germany.

This project was funded by grants from the Werner Siemens Foundation (Paleobiotechnology to C.W. and P.S.) and the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation, under Germany's Excellence Strategy – EXC 2051 – Project-ID 390713860 to C.W. and P.S.). J.A.F.Y and C.W. were funded by the Deutsche Forschungsgemeinschaft (DFG, German Research Foundation) – project number 460129525 (NFDI4Microbiota, FlexFund project EnterArchaeo). J.A.F.Y and C.W. were supported by the Max Planck Society. O.C.D. and A.E.M. were supported with computing infrastructure access by the West University of Timisoara (hpc.uvt.ro), acquired via grant no. 240/2020, ID 911 POC/398/1/1, financed by the European structural funds and Romanian government funds, and the project "Romanian Hub for Artificial Intelligence - HRIA", in the program PCID-IF/709/PCIDIF_P4/OP1/RSO1.6/PCIDIF_A12 - Action 4.1 - MySMIS 351416. This work was supported by the de.NBI Cloud within the German Network for Bioinformatics Infrastructure (de.NBI) and ELIXIR-DE (Forschungszentrum Jülich and W-de.NBI-001, W-de.NBI-004, W-de.NBI-008, W-de.NBI-010, W-de.NBI-013, W-de.NBI-014, W-de.NBI-016, W-de.NBI-022).

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